

3GPP TS 24.501

Non-Access-Stratum (NAS) protocol for 5G System (5GS)

Version 17.8.0 Release 17

SYNTHETIC SAMPLE DOCUMENT — written for demonstration and testing of a retrieval-augmented chatbot. This is original content structured to resemble a 3GPP technical specification; it is NOT an official 3GPP publication and must not be used as a source of normative requirements.

5.1.3 Registration procedure

The registration procedure is used by the UE to register with the network for initial registration, mobility registration update, periodic registration update, or emergency registration. The UE sends a REGISTRATION REQUEST message to the AMF. If authentication and security procedures complete successfully and the AMF accepts the request, it responds with a REGISTRATION ACCEPT message, which the UE acknowledges with REGISTRATION COMPLETE if new 5G-GUTI or TAI list values were assigned.

5.5.1 De-registration procedure

De-registration may be UE-initiated or network-initiated. In UE-initiated de-registration, the UE sends a DEREGISTRATION REQUEST indicating whether it wishes to switch off. In network-initiated de-registration, the AMF sends a DEREGISTRATION REQUEST to the UE, which shall respond with DEREGISTRATION ACCEPT unless the switch-off indicator makes an acknowledgement unnecessary.

10.2 NAS timers

The following timers control retransmission and supervision of key NAS procedures at the UE side. Timer T3510 supervises the registration procedure; if it expires before REGISTRATION ACCEPT or REGISTRATION REJECT is received, the UE retransmits REGISTRATION REQUEST up to a configured number of times. Timer T3521 supervises the de-registration procedure in an analogous way.

Timer	Default value	Retransmission limit
T3510	15 seconds	5 attempts
T3521	15 seconds	5 attempts
T3511	10 seconds	N/A — cell re-selection timer

5.4.1 Authentication procedure

The network may initiate NAS authentication at any time by sending an AUTHENTICATION REQUEST containing an authentication challenge. The UE calculates a response using its USIM and returns it in AUTHENTICATION RESPONSE. If the response does not match the expected value, the network sends AUTHENTICATION REJECT and the UE shall consider the USIM invalid for 5GS services until the UE is switched off or the USIM is removed.